Inspecting every task and document length

Paired gains over vanilla synthetic data · percentage points · all 40 task–length groups retained

a LESS DILUTION

39 / 40 groups improve

	8K	16K	32K	64K
QA1	+22.2	+36.2	+40.0	+33.1
QA2	+26.7	+45.1	+44.5	+36.7
QA3	-2.0	+11.2	+17.8	+27.1
QA4	+36.6	+39.5	+28.5	+21.5
QA5	+18.1	+27.5	+39.3	+37.2
QA6	+22.0	+32.3	+37.4	+41.4
QA7	+31.8	+52.6	+54.9	+63.0
QA8	+22.8	+37.0	+37.3	+47.2
QA9	+32.8	+34.4	+33.4	+38.0
QA10	+31.4	+30.1	+37.9	+34.2

Vanilla − query-specific

Zero-support context share

b HIGHER COVERAGE

38 / 40 groups improve

	8K	16K	32K	64K
QA1	+28.0	+44.0	+76.0	+72.0
QA2	+12.0	+50.0	+44.0	+54.0
QA3	-13.9	+13.3	+32.0	+49.3
QA4	+44.0	+52.0	+54.2	+72.0
QA5	0.0	+20.0	+60.0	+72.0
QA6	+24.0	+40.0	+64.0	+88.0
QA7	+9.5	+33.8	+59.9	+74.5
QA8	+11.3	+43.3	+64.5	+86.0
QA9	+20.0	+32.0	+52.0	+84.0
QA10	+20.0	+28.0	+64.0	+76.0

Query-specific − vanilla

Dataset support-fact recall

c MORE JOINT EXPOSURE

15 / 16 groups improve

	8K	16K	32K	64K
QA1	—	—	—	—
QA2	+13.1	+18.5	+21.5	+10.7
QA3	-0.4	+7.5	+6.0	+5.5
QA4	—	—	—	—
QA5	—	—	—	—
QA6	—	—	—	—
QA7	+8.4	+15.7	+20.5	+18.8
QA8	+21.4	+13.2	+13.8	+14.6
QA9	—	—	—	—
QA10	—	—	—	—

Query-specific − vanilla

All-fact co-exposure; multi-fact cases

Worse ← −100 −50 0 50 100 → Better

Each cell averages its paired cases; maximum n = 25. “—” = no multi-fact cases, not zero gain. Exact counts are in the source table.